layer, which is expected to capture the direction in representation space most associated with self-reflection. Specifically, we modify each residual stream $\boldsymbol{h}^{(\ell)}$ at the ℓ-th layer in (1) according to

$$\widehat{\boldsymbol{h}}^{(\ell)} = \boldsymbol{h}^{(\ell)} + \alpha \boldsymbol{v}^{(\ell)} \left\langle \boldsymbol{h}^{(\ell)}, \boldsymbol{v}^{(\ell)} \right\rangle, \tag{6}$$

where $\widehat{\boldsymbol{h}}^{(\ell)}$ then replaces $\boldsymbol{h}^{(\ell)}$ as the input to the next layer, and the scalar α controls the strength of the intervention. When $\alpha > 0$, the model's self-reflection behavior is enhanced; when $\alpha < 0$, it is suppressed. Setting $\alpha = 0$ disables the intervention, preserving the model's default behavior.

Ablation study on α To illustrate the effect of the linear intervention method for controlling self-reflection, we conduct an ablation study by varying the self-reflection steering strength α from -1.0 to 1.0, injecting the reflection vector at layer 14 in DeepSeek-R1-1.5B on the MATH-500 benchmark. The result is shown in Figure 4. Negative α values shorten responses, reducing average token length, while preserving accuracy. In contrast, positive α both lengthens responses, indicating deeper self-reflective reasoning, and boosts performance, peaking at α=0.03 with a 12% performance gain in Pass@1, before declining at larger values due to over-reflection. This clear trade-off underscores α as a practical knob for balancing verbosity against reasoning depth. Further ablation detail on the effect of the injection layer is provided in Appendix C.

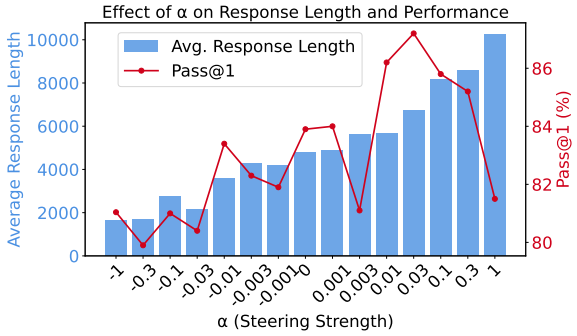


Figure 4: Effect of α on performance and response length on Math500 Dataset (reflection vector is injected on layer 14).

4.3 Experimental Results

We evaluate our self-reflection control mechanism on two mathematical reasoning benchmarks, MATH-500 [29] and AIME 2024, and one scientific QA benchmark, GPQA Diamond [34]. Experiments are conducted using DeepSeek-R1 and Qwen2.5 models at both 1.5B and 7B parameter scales, as well as the Llama 3.1 8B Instruct model [7,30,35].

We compare three inference strategies: **Vanilla (Baseline)**, which uses standard setting without any intervention; **BF (Budget Forcing)**, which enforces reflection by appending a "wait" token at the end of initial short generations [7,36]; and **Self-Reflection (SR) Enhanced/Suppressed**, our proposed technique that perturbs hidden states using self-reflection vectors scaled by a coefficient α (positive for enhancement, negative for suppression). For details on selecting the optimal injection strategy, please refer to Appendix C.

Key Findings. Our results highlight three major insights. First, SR Enhancement improves reasoning performance across most evaluated datasets and model sizes. For example, Qwen2.5 7B's performance on MATH-500 jumps by 12.0 percentage points (from 44.6% to 56.6%), and DeepSeek-R1 variants enjoy similar boosts when employing reflection-enhanced decoding. Furthermore, this performance gain is also accompanied by a noticeable increase in response length, suggesting that longer responses can be beneficial when tackling more challenging reasoning tasks.

Second, SR Suppression offers fine-grained control over computational cost. It consistently reduces output length—often by more than 50%—while preserving most of the model's accuracy. Notably, DeepSeek-R1-7B reduces average token length from 3564 to 2451 on MATH-500 with only a minor drop in Pass@1, which remains above 91%.

Finally, these effects demonstrate strong generalizability across training paradigms and model families. The observed improvements hold for both $\mathcal{A}_{\texttt{ft}}$ models (e.g., DeepSeek-R1) and $\mathcal{A}_{\texttt{pt}}$ models (e.g., Qwen2.5, Llama3.1), suggesting that the self-reflection signal is a robust and transferable mechanism that transcends specific architectures or fine-tuning methods.

Size	Method	MATH-500		AIME 2024		GPQA Diamond	
		Pass@1 ↑	LEN ↓	Pass@1 ↑	LEN ↓	Pass@1 ↑	LEN ↓
DeepSeek-R1-1.5B	Vanilla	84.1 ± 0.4	4755 ± 216	29.2 ± 1.1	6118 ± 351	14.0 ± 0.9	4250 ± 285
	BF	85.5 ± 0.5	10122 ± 452	30.0 ± 1.3	8986 ± 410	14.8 ± 0.7	5210 ± 339
	SR Enhanced	**87.4 ± 0.3**	9458 ± 424	**33.5 ± 1.2**	8132 ± 392	**18.9 ± 0.8**	7496 ± 462
	SR Suppressed	83.2 ± 0.6	**3716 ± 151**	27.3 ± 1.4	**5229 ± 248**	13.8 ± 1.0	**3795 ± 199**
DeepSeek-R1-7B	Vanilla	92.7 ± 0.3	3585 ± 181	55.8 ± 2.0	4558 ± 253	27.1 ± 1.6	3696 ± 215
	BF	93.1 ± 0.2	7111 ± 319	54.5 ± 2.4	9629 ± 488	32.0 ± 1.7	5802 ± 347
	SR Enhanced	**93.5 ± 0.3**	8959 ± 402	**58.2 ± 1.9**	5684 ± 308	**34.6 ± 1.5**	6513 ± 382
	SR Suppressed	91.2 ± 0.4	**2439 ± 118**	52.9 ± 2.2	**3319 ± 162**	26.5 ± 1.8	**3120 ± 156**
Qwen2.5 1.5B	Vanilla	27.5 ± 1.5	**1515 ± 98**	0.2 ± 0.1	**544 ± 58**	5.7 ± 0.8	**702 ± 67**
	BF	29.6 ± 1.8	3993 ± 252	0.1 ± 0.1	4517 ± 305	5.0 ± 0.9	1389 ± 114
	SR Enhanced	**36.9 ± 1.4**	1836 ± 129	**6.1 ± 0.9**	2525 ± 180	**6.8 ± 0.7**	814 ± 70
Qwen2.5 7B	Vanilla	44.8 ± 2.1	**1294 ± 88**	3.9 ± 0.8	**2285 ± 153**	16.2 ± 1.4	**598 ± 40**
	BF	46.0 ± 1.9	3986 ± 242	5.3 ± 0.9	4526 ± 296	**16.9 ± 1.3**	1639 ± 135
	SR Enhanced	**56.8 ± 1.7**	2671 ± 174	**16.1 ± 1.4**	2941 ± 202	14.8 ± 1.5	1816 ± 147
Llama 3.1 8B Instruct	Vanilla	44.0 ± 2.0	**1613 ± 112**	3.5 ± 0.6	**1653 ± 125**	30.9 ± 1.9	**1914 ± 144**
	BF	40.1 ± 2.0	3771 ± 266	2.9 ± 0.7	3912 ± 284	26.1 ± 2.4	5031 ± 351
	SR Enhanced	**57.7 ± 1.6**	2887 ± 198	**16.5 ± 1.3**	2974 ± 210	**34.1 ± 1.8**	3995 ± 310

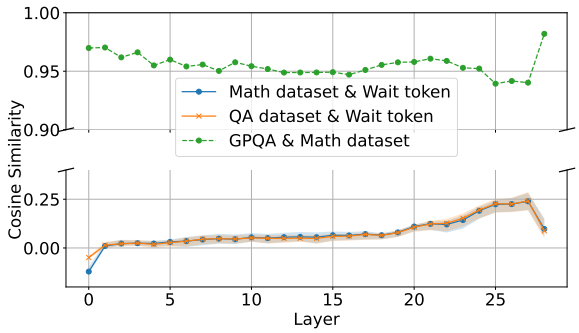
Table 1: Performance across mathematical and scientific reasoning benchmarks using models of different sizes. We compare three inference strategies: **Vanilla** (no intervention), **BF** (budget forcing via "wait" token insertion), and our method: **SR Enhanced/Suppressed** (applying positive or negative α to respectively amplify or suppress self-reflection during inference). Pass@1 indicates accuracy (higher is better); LEN indicates average generation length (lower is better). We use 10% of the data as a validation set to select the optimal α in Eq. 6, and report test performance averaged over five runs with different random seeds.

Building on this generalizability, we emphasize the practical value of inference-time control over latent self-reflection dynamics. Unlike rigid interventions such as budget forcing, our method affords semantically grounded, continuous modulation of a model's internal self-reflection, enabling a tunable trade-off between performance and efficiency, especially in resource-constrained settings.

4.4 Transferability of Self-Reflection Vectors

To investigate the transferability of self-reflection vectors across different reasoning domains, we evaluated whether vectors extracted from the GPQA Diamond dataset could be effectively transferred to mathematical reasoning tasks in MATH500. We compute the cosine similarity between self-reflection vectors extracted from different domains (MATH500 and GPQA Diamond), and between these vectors and the embedding of the token "wait" in DeepSeek-R1-Distill-Qwen-1.5B. For tokenizers containing multiple subword tokens for "wait", we report the average cosine similarity along with its variance. The results are plotted in Figure 5(left). Our analysis revealed remarkable consistency in the neural signatures of self-reflection across these distinct domains. Specifically, vectors extracted from GPQA and MATH500 exhibit high cosine similarity, suggesting that the internal representation of reflective states is largely domain-invariant. Notably, these self-reflection vectors are substantially different from the embedding of the token "wait", indicating that they encode deeper semantic properties of reflective behavior rather than surface-level cues.

Further, we evaluate the performance of self-reflection vectors derived from GPQA-Diamond on MATH500 using our proposed intervention method, SR-Enhanced/Suppressed, and present the results in Figure 5(right). Notably, we observe similar performance gains to those seen with in-domain self-reflection vectors, as reported in Table 1. This cross-domain transfer demonstrates that the reflective mechanism captures a generalizable cognitive pattern rather than being confined to task-specific reasoning strategies. Together, these findings suggest that LLMs develop a unified internal representation of self-reflection, one that can be leveraged across tasks without the need for domain-specific fine-tuning.



Model	Setting	PASS@1	Len
1.5B	SR Enhanced	86.4	7781
	SR Suppressed	82.6	3684
7B	SR Enhanced	93.0	6007
	SR Suppressed	91.0	2992

Figure 5: **Left:** Cosine similarity of self-reflection vectors and the "wait" token across MATH500 and GPQA datasets. The green curve shows similarity between vectors from MATH500 and GPQA. Blue and orange curves show similarity with the "wait" token. **Right:** Performance on MATH500 when applying self-reflection vectors extracted from GPQA Diamond to DeepSeek-R1 models.

5 Related Work

5.1 Features as Directions

Extracting feature directions, often derived from contrastive pairs of inputs, is an established technique for analyzing and manipulating neural network representations [16,37,38]. It is widely recognized that adding such feature vectors to the model's residual stream can modify its behavior, although the optimal intervention points and specific methodologies remain areas of active research [39,40].

Several studies suggest that directions within the activation space capture semantic features more effectively or interpretably than individual neurons [41–43]. Recent approaches utilize techniques like sparse autoencoders to discover these feature directions in an supervised manner [44]. However, alternative methods such as Difference-in-Means (DiffMean) [32] have demonstrated strong performance, sometimes exceeding that of sparse autoencoders, in specific applications like concept detection and model steering [33]. Furthermore, the underlying assumption that features can be represented linearly has proven effective in tasks such as targeted concept erasure within language models [45–47].

5.2 Self-Reflection in Language Models

The concept of self-reflection in language models has gained increasing attention as a mechanism for improving reasoning quality and alignment. Recent studies [48–53] have explored how prompting models to generate intermediate reflections, critiques, or alternative solutions can improve final outputs in tasks such as math problem solving, programming, and factual reasoning. While many of these techniques are implemented at the prompting level or through chain-of-thought scaffolding, they suggest that self-reflection is a powerful tool for enhancing reasoning. Notably, these methods often induce substantial increases in generation length and latency, raising questions about the trade-off between deliberation and efficiency [13,54–56].

Recent studies have show that RLVR [1,57–59] can improve reasoning abilities by explicitly training models to reflect using outcome-based feedback. Empirically, trained models [7,60] like DeepSeek-R1 demonstrate significant improvements over baseline models in mathematical and logical reasoning tasks, and showcase new emergent behaviors such as self-reflection. Our work shows that self-reflection is a broadly distributed and latent feature of LLMs, not exclusively a product of RLVR. Concurrent works [11,57] also suggest that RLVR does not necessarily introduce novel reasoning abilities beyond those acquired during pretraining; instead, it primarily serves to amplify abilities already present in the model. Our work also complements this literature by showing that LLMs already encode latent self-reflection signals in their hidden states—even in models not explicitly trained for such behavior—and that reflection can be selectively enhanced or suppressed through lightweight vector interventions. This enables fine-grained control over reflective